BTS-VII-11.16-1120 Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

D

B.Tech. Degree VII Semester Examination November 2016

CS 1704 ANALYSIS AND DESIGN OF ALGORITHMS

(2012 Scheme)

Time : 3 Hours Maximum Marks : 100

PART A
(Answer *ALL* questions)

$(8 \times 5 = 40)$

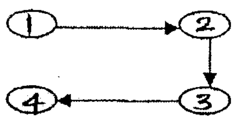
I. (a) What do you understand by analysis of algorithms? Briefly describe various criteria for analyzing algorithms.

(b) Give a brief description on dynamic programming.

(c) Explain heap sort procedure with elements 29, 30, 8, 39, 5.

(d) What is amortized time analysis?

(e) Perform Warshall's algorithm for transitive closure on the following digraph.



(f) Write notes on:

(i) Articulation point (ii) Biconnected graph.

(g) Briefly describe one greedy strategy for TSP with an example.

(h) What is meant by optimization problem?

SCHOOL OF ENGINEERING, KOCHI-682 022 * LIBRARY * COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

PART B

$(4 \times 15 = 60)$

II. Solve: (15)

(i) $T(n) = 2T(n/2) + n\lg n$
(ii) $T(n) = 2T(\sqrt{n}) + \lg n$
(iii) $T(n) = 2T(n/2) + \sqrt{n}$

OR

III. Explain the different asymptotic notations used for specifying the growth rate of functions. (15)

IV. (a) Write an algorithm for insertion sort. Analyze its complexity. (10)
(b) Compare any two sorting algorithms. (5)

OR

V. (a) Explain worst case and best case complexity of quick sort with example. (8)
(b) Explain insertion operation in Red-Black Trees. (7)

(P.T.O.)

VI.	(a)	Discuss biconnected component of an undirected graph.	(5)
	(b)	What is strongly connected component of a digraph? Explain algorithm with an example.	(10)

OR

VII.		Explain anyone algorithm for finding all pair shortest path in graphs with example.	(15)
VIII.	(a)	What is bin packing? Explain different strategies for bin packing with example.	(12)
	(b)	Explain class P.	(3)

OR

IX.	(a)	Distinguish between NP Hard and NP complete problem.	(10)
	(b)	Write a short note on approximation algorithms.	(5)

***